COMPOSITE MATERIAL DATASHEETS CFRP HiTape/RTM6-UD (V800)

MATERIAL IDENTIFICATION

Short informal name	HiTape/RTM6-UD_V800
Commercial name	Hexcel HiTape 210gsm/IMA/2xV800E 4g/m2 with RTM6
Specifications	(*) This material is not qualified according Airbus standards and specifications. The use of the design data presented in this report is restricted to the research project in which it has been used, i.e. the *Clean Sky 2*, . in a demonstrator of an external wing structure.
Description	Carbon fiber uni-directional fabric in epoxy resin. Produced by a liquid resin infusion (LRI) process. Service temperature: -55 to 90ºC. Cure: 120-135 minutes at 185 +-5ºC.
Version	0.7 (26/07/2022)

REFERENCES

- TAE-CS-NT-160016 Is. D “ANALYSIS OF HEXCEL HiTape / RTM6 COUPONS TEST RESULTS AND ALLOWABLE VALUES DERIVATION”
- TEA-CS-NT-170061 Is. A “HiTape/RTM6 (LRI) Compression After Impact CAI. TEST RESULTS ANALYSIS.”
- TEA-CS-NT-180110 Is. B “RESIDUAL STRENGTH AFTER IMPACT OF CFRP HEXCEL HITAPE/RTM6 UD-FABRIC LAMINATES”
- [Hexcel] TS-AR 18112001 “Measurement of coefficient of thermal expansion of HiTape®/ RTM6 laminates (quasi-isotropic, 0° & 90°)”
- ESCRIBIR ACTUALIZACION DE LA TEA-CS-NT-180110 CON COMALL 0.22 ETC…

PHYSICAL PROPERTIES

Cured Ply Thickness (CPT)	0.196	mm
Density		kg/m³
Tg-onset (wet)		ºC
Max. Operational Limit (MOL)		ºC

Coefficients of Thermal Expansion (1/ºC)	Range	-55..20ºC	20..70ºC	70..120ºC
	α_1	0.0	0.0	
	α_2	3.1E-5	3.5E-5	
	α_3	3.1E-5	3.5E-5	

ELASTIC MODULI

Property	E_1	E_2	G_{12}	ν_{12}	K_B
Units	MPa	MPa	MPa	-	-
Values	160000	8500	3800	0.32	0.8

Property	E_3	G_{13}	G_{23}	ν_{13}	ν_{23}
Units	MPa	MPa	MPa	-	-
Values	8500	3800	2700	0.32	0.586

Remarks:

- The K_B is the buckling correction factor for moduli to apply to E_1, E_2 and G_{12} in buckling stability analyses

PLAIN STRENGTHS

Ultimate stresses of the ply (from UD-laminates testing).

Property	Mean Values (MPa)				B-basis	E-KDF			B-basis Values (MPa)			
	RT/Dry	70ºC/W	90ºC/W	120ºC/W	K_B	70ºC/W	90ºC/W	120ºC/W	RT/Dry	70ºC/W	90ºC/W	120ºC/W
F_{1t}	2816	2221	-	-	0.781	0.788	-	-	2200	1734	-	-
F_{1c}	1254	1010	-	-	0.734	0.805	-	-	920.7	741.2	-	-
F_{2t}	47.1	30.3	-	-	0.779	0.643	-	-	36.7	23.6	-	-
F_{2c}	191.3	134.6	-	-	0.829	0.704	-	-	158.6	111.6	-	-
F_{12}	96.7	71	-	-	0.845	0.734	-	-	81.7	60	-	-

Design values for checking the laminate strength using the modified Maximum Strain Criterion.

Property	Mean Values (με)				B-basis	E-KDF			B-basis Values (με)			
	RT/Dry	70ºC/W	90ºC/W	120ºC/W	K_B	70ºC/W	90ºC/W	120ºC/W	RT/Dry	70ºC/W	90ºC/W	120ºC/W
ε_{tall}	16800	13900	-	-	0.777	0.827	-	-	13100	10800	-	-
ε_{call}	7800	6300	-	-	0.734	0.808	-	-	5700	4600	-	-

INTERLAMINAR PLY STRENGTHS

Property	Mean Values				B-basis	E-KDF			B-basis Values			
	RT/Dry	70ºC/W	90ºC/W	120ºC/W	K_B	70ºC/W	90ºC/W	120ºC/W	RT/Dry	70ºC/W	90ºC/W	120ºC/W
F_{13} (MPa)	75.3	58.2	-	-	0.887	0.773	-	-	66.8	51.6	-	-
F_{3t} (MPa)	-	-	-	-	-	-	-	-	-	-	-	-
F_{3c} (MPa)	-	-	-	-	-	-	-	-	-	-	-	-
G_{IC} (N/mm)	1.015	1.541	-	-	0.83	1.0	-	-	0.842	0.842	-	-
G_{IIC} (N/mm)	-	-	-	-	-	-	-	-	-	-	-	-

Notes: F3t shall be made equal to F2t in QI and directed laminates if thickness ≥3mm, otherwise contact Stress Methods dpt. Through-thickness compression strength F3c shall be made equal to F2c

OPEN HOLE AND BOLTED JOINTS

Reference material strengths in open-hole, filled-hole and bearing for the quasi-isotropic laminate.

Property	Mean Values (MPa)				B-basis	E-KDF			B-basis Values (MPa)			
	RT/Dry	70ºC/W	90ºC/W	120ºC/W	K_B	70ºC/W	90ºC/W	120ºC/W	RT/Dry	70ºC/W	90ºC/W	120ºC/W
OHT_{QI}	581	-	-	-	0.9	0.94	-	-	522.9	491.8	-	-
OHC_{QI}	270.6	212.2	-	-	0.881	0.784	-	-	238.4	187	-	-
FHT_{QI}	555.5	-	-	-	0.891	0.94	-	-	494.9	465.5	-	-
FHC_{QI}	389.8	320.5	-	-	0.824	0.822	-	-	321.3	264.2	-	-
$F_{BR\text{-}QI}$	950.5	822.5	-	-	0.874	0.865	-	-	831.2	719.2	-	-
F_{PO}	-	-	-	-	-	-	-	-	-	-	-	-

Note: EKDF's for OH and FH strengths taken from plain-strength data

COMPOSITE DAMAGE TOLERANCE

- Residual strength after transverse impact

  The following TAI, CAI model is valid for laminates following the standard design rules (symmetric, balanced, with minimum 10% of plies at angles 0º, 90º, 45º and -45º).

Property	B-basis Value for all conditions
TAI (με)	7000

Note: TAI estimated as 0.8·OHT, and EKDF taken from PTS data.

$$E/t^2 = A\cdot(1 - e^{-B\cdot d0/t})$$

Approximation model type: CDTM-T07

$$CAI = K_B \cdot EKDF \cdot CAI_{\infty\text{-}QI} \cdot (3 / K_{TOH})^C \cdot [\ 1 + A / e^{B\cdot(E/t\wedge 2)}\ ]$$

Property	Design Value
A (J/mm²)	5.4
B	8.0

Property	Mean Value RT/Dry	B-basis K_B	E-KDF 70ºC/W	E-KDF 90ºC/W	E-KDF 120ºC/W
$CAI_{\infty\text{-}QI}$ (με)	3000	0.83	0.85	-	-
A	1.25				
B (mm²/J)	0.36				
C	1.45				

…where:

- *E* is the impact energy, not higher than the cut-off identified during damage threats analysis (for many parts of the airframe, 35J as cut-off value shall be acceptable)
- *t* the laminate thickness
- d_0 the dent depth after impact (typ. 2.0-2.5mm for BVID detectability under a GVI inspection)
- K_{TOH} is the Stress Concentration Factor for an infinite plate with a circular hole in by-pass load, calculated from laminate apparent moduli for X direction (*) as:

$$K_{TOH} = 1 + \{ 2\cdot[(E_x/E_y)^{0.5} - \nu_{xy}] + E_x/G_{xy} \}^{0.5}$$

(*) CAI shall be calculated not only for the 0º direction (X parallel to material 0º reference), but also for the rest of ply orientations, i.e. the 45º and 90º directions, by rotating the X axis and re-evaluating previous expressions with the new values of E_x, E_y, G_{xy} and ν_{xy} . In a general case CAI strength shall be expressed as 3 scalar values with the alowable strains applicable to plies at 0º (CAI_0), at ±45º (CAI_{45}) and at 90º (CAI_{90}).

- Residual strength after transverse BVID near lightening holes

Property	B-basis value up to 70ºC/W
CAI_{hole} (με)	7600

…to be compared with the peak tangential strain calculated at the edge of a hole in a panel made of this material. Source: tests on impacted holed panels made of Hexcel HiTape/RTM6 UD-Tape.

- Residual strength after edge impact

No design data for *CAIE* still available

Warning! These design values shall be considered preliminary and not used for structural justification assessments in certification/qualification processes (this material is still not qualified).